

# Fixed-point $Q_m.n$

Hardware often needs fractions without a full floating-point unit

# Scale, step, and width

- In this lab, total width  $W$  equals  $m$  plus  $n$
- The smallest positive step is two to the minus  $n$
- Encoding a real multiplies by two to the  $n$  and rounds into the bit field
- Changing  $n$  with the same raw bits moves the binary point and changes the real value
- $M$  includes the sign bit here

# Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Qm.n

INTERACTIVE TOOL

Fixed-point Qm.n

See how **integer** and **fractional** bits share one word — encode/decode with scale  $2^{-n}$  and a binary point.

Starter example: Q4.4 (8 bits). Pattern 0001\_1000 is raw 24 → real 1.5 ( $24 / 2^4$ ). Move the binary point by changing n.

Load starter example

Challenges 0 / 22 cleared

Quiz: scale: Decode divides signed raw by? Answer:  $2^n$

Answer

Show hint

Check

Next

Idle

Quiz: scale

Quiz: step

Quiz: width

Quiz: m includes

Starter 1.5

Decode

Encode 0.5

Encode -1.25

Saturate

Change n

Quiz: Q1.15

Set Q1.15

Binary point

Quiz: raw 1.5

Encode 0

Toggle LSB

Quiz: Q4.4 min

n=0 integer

Round-trip

Quiz: notation

Q5.4

Full Q story

Core ideas

Qm.n

m integer bits (incl. sign) + n fraction bits. Scale  $2^{-(n)}$ .

Decode

Real = signed raw integer ÷  $2^n$ . Encode multiplies then rounds.

Fixed-point starter

learn\_digital — fixed point

# Workbook practice

- In the workbook track, take Q four-dot-four
- Show why raw twenty-four is one-point-five, and why raw eight is zero-point-five
- Compute the step size as one sixteenth
- Note the approximate min near minus eight
- Name one pitfall: treating raw bits as a float without dividing by two to the n

# Pitfalls to watch

- Do not forget the scale when you mix formats
- Saturation is not the same as wrap
- And remember: the browser lab is literacy
- Real DSP still needs agreed Q formats, rounding modes, and overflow policy across blocks

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, encode and decode one Q four-dot-four value by hand
- When you are ready, take the short quiz, then continue to bit-fields